

Jeremy Malloch

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Skills

Python, PyTorch, ML Modelling, ML Deployment

Work Experience

Machine Learning Engineer – Cruise *San Francisco, CA* July 2021 – Present

- Owned overall release of second iteration of model (intended for Cruise's return to driverless), saw net 13 major progressions in ultra nearfield (UNF) suite (target of release)
 - Coordinated with upstream model retrains, new features, and dataset additions
 - Integrated multiple UNF datasets, resulting in ~33% increase in dataset size
 - Released on schedule, before driverless code freeze
- Contributed multiple modelling improvements to model:
 - Increased MFU by **32%**
 - Stabilized training by analyzing model activation magnitudes
 - 20% reduction in velocity flips
 - 11% reduction in centroid error, 12% reduction in velocity error, 3.25x as many major progressions vs regressions
 - Increased recall of pedestrians from **95.03% to 98.33%**, with improved F1 scores in 5/6 semantic classes
- Created model attention metrics, directly leading to modeling improvements
- Owned deployment of novel tracking ML model, achieved 3x speedup of node, coordinated with downstream consumers to resolve latency blockers
- Developed in house PyTorch optimization library (quantization, pruning) with **torch.fx**, used to decrease inference latency of multiple deployed models.
- Lead partnership to build compiler frontend / transformation for bit accurate emulation of DNN inference on future hardware
- Collaborated with open source, contributing bug fixes and reporting issues to TensorRT, PyTorch, and Netron

Autonomous Systems Software Intern - Apple *Sunnyvale, CA* Jan 2019 – Aug 2019

- Contributed to machine learning compiler/inference framework in **C++ & Python**
- Researched & prototyped DNN quantization & sparsification pipeline, resulting in up to **40% faster** runtime of deployed models

Machine Learning Intern - Cognitive Systems *Waterloo, ON* May 2018 – Aug 2018

- Achieved **60x** improvement in neural network inference runtime by implementing im2col convolutions, vectorization and memory alignment with Eigen in **C++** for ARM CPU (exceeded **Tensorflow Lite** baseline by **2x**)
- Researched & prototyped semi-supervised GAN data augmentation to reduce false positives

Projects

Semi-supervised Image Representation Model Nov 2017 – Jan 2018

- Trained a ResNet to learn feature representations by predicting layout of 3x3 image subcrops, inspired by research paper (<https://arxiv.org/abs/1603.09246>)

Education

University of Waterloo

Waterloo, ON

BMath in Computer Science and Combinatorics & Optimization

2021

Relevant Courses: Algorithms, Operating Systems, Real Analysis, Group Theory, Mathematical Optimization (Convex & Linear Programming), Deep Learning in Discrete Optimization (graduate)